The Harmonic Range and Maximum Entropy

A Projective-Geometric Characterization of the Logistic Map at $r = 4$

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ABSTRACT

The logistic map $f_r(x) = rx(1-x)$ at parameter $r = 4$ is distinguished by a well-known constellation of dynamical properties: surjectivity on $[0, 1]$, maximal topological entropy $\log 2$, an absolutely continuous invariant measure, and postcritical finiteness. We observe that these properties are equivalent to a classical condition in projective geometry: the four structurally distinguished points of f_r on $\mathbb{CP}^1$ — the critical point, critical value, non-trivial fixed point, and the point at infinity — form a *harmonic range* (cross-ratio -1) if and only if $r = 4$. This characterization emerges naturally from the realization of f_4 as a quotient of the doubling map on the circle by complex conjugation, whose symmetry group is the Klein four-group $V_4 \cong \mathbb{Z}/2 \times \mathbb{Z}/2$. The same group is the universal stabilizer of the cross-ratio in S_4 and underlies the enhanced D_4 symmetry specific to the harmonic value $\lambda = -1$. The harmonic condition is moreover equivalent to f_4 being the unique non-trivial Belyi map in the logistic family — its critical values lie in $\{0, 1, \infty\}$ — connecting the dynamical characterization to arithmetic geometry and the moduli space $\mathscr{M}_{0,4}$.

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1. Introduction

The logistic family $f_r(x) = rx(1-x)$, $r \in [0, 4]$, is among the most studied objects in dynamical systems. At the endpoint $r = 4$, the map achieves maximum complexity in a precise sense: it is semiconjugate to the full one-sided shift on two symbols, its topological entropy equals $\log 2$, and it admits an absolutely continuous invariant measure — the arcsine distribution. These facts have been understood since at least the work of Ulam and von Neumann [1]. For modern treatments, see [2] and [3].

The purpose of this article is to observe a projective-geometric condition that belongs naturally on this list. The four dynamically distinguished points of f_r on the Riemann sphere $\mathbb{CP}^1$ — the critical point c, the critical value v, the non-trivial fixed point p, and the point ∞ — form a **harmonic range** (cross-ratio equal to -1) if and only if $r = 4$. This condition is equivalent to the statement that p is the affine midpoint of c and v, or equivalently, that the unique Möbius involution fixing p and ∞ sends c to v.

The harmonic range is the most symmetric non-degenerate configuration of four points on a projective line, with an enhanced stabilizer under permutation: the dihedral group D_4 of order 8, which contains the Klein four-group $V_4 = \mathbb{Z}/2 \times \mathbb{Z}/2$ as a normal subgroup. This same V_4 appears independently as the symmetry group of the doubling map $z \mapsto z^2$ on the unit circle, whose quotient by complex conjugation yields f_4. Both manifestations of V_4 trace to the degree-2 structure of the underlying map.

We further observe that the harmonic condition is equivalent to f_4 being a *Belyi map* — a rational map whose critical values lie in $\{0, 1, \infty\}$. This places the characterization in the setting of arithmetic geometry: by Belyi's theorem, such maps correspond to algebraic curves defined over number fields, and their combinatorial data (dessins d'enfants) carry an action of the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. The postcritical set $\mathrm{PC}(f_4) = \{0, 1, \infty\}$ coincides with the boundary divisor $\partial\overline{\mathscr{M}}_{0,4}$ of the Deligne–Mumford compactification of the moduli space of four marked points on $\mathbb{CP}^1$, a relationship that the Belyi perspective clarifies.

The article is organized as follows. Section 2 recalls the semiconjugacy between the doubling map and f_4 via the quotient by complex conjugation. Section 3 identifies the full Klein four-group of symmetries and its quotients. Section 4 defines the four structural points, computes the cross-ratio, and proves the harmonic range characterization. Section 5 connects the harmonic condition to the Belyi property and the moduli space $\mathscr{M}_{0,4}$. Section 6 discusses entropy, surjectivity, and the variational principle. Section 7 synthesizes the full equivalence and offers concluding remarks.

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2. The Doubling Map and Its Quotient

2.1. The doubling map on S^1

On the unit circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, the **doubling map** is

$$D : S^1 \to S^1, \qquad D(z) = z^2,$$

or equivalently $D\left(e^{i\theta}\right) = e^{2i\theta}$. This is the fundamental chaotic system of degree 2:

- Every point has exactly two preimages (z and $-z$).
- Lebesgue measure $\frac{d\theta}{2\pi}$ on S^1 is invariant and ergodic.
- The topological entropy is $h_{\mathrm{top}}(D) = \log 2$.

- The symbolic dynamics, via the partition $S^1 = \left\{e^{i\theta} : \theta \in [0, \pi)\right\} \cup \left\{e^{i\theta} : \theta \in [\pi, 2\pi)\right\}$, is the full one-sided shift on $\{0, 1\}^{\mathbb{N}}$.

2.2. The quotient by complex conjugation

Complex conjugation restricts to an involution on S^1:

$$\beta : S^1 \to S^1, \qquad \beta(z) = \bar{z}.$$

Since $|z| = 1$ implies $\bar{z} = z^{-1}$, this identifies $e^{i\theta}$ with $e^{-i\theta}$. Crucially, β **commutes** with D:

$$D(\bar{z}) = \bar{z}^2 = \overline{z^2} = \overline{D(z)}.$$

The quotient $S^1/\langle\beta\rangle$ (where $\langle\beta\rangle = \{e, \beta\}$ denotes the subgroup generated by β) is homeomorphic to $[0, \pi]$ via $e^{i\theta} \mapsto \theta \in [0, \pi]$, and the doubling map descends to a map on this interval. Under the change of variable

$$x = \sin^2\left(\frac{\theta}{2}\right) = \frac{1 - \cos\theta}{2},$$

which maps $[0, \pi]$ homeomorphically onto $[0, 1]$, the descended map takes the form

$$f_4(x) = 4x(1 - x).$$

Verification. Starting from $x = \frac{1-\cos\theta}{2}$:

$$f_4(x) = 4 \cdot \frac{1 - \cos\theta}{2} \cdot \frac{1 + \cos\theta}{2} = 1 - \cos^2\theta = \sin^2\theta = \frac{1 - \cos 2\theta}{2}.$$

This is the value of x at 2θ, confirming that f_4 is the descent of $\theta \mapsto 2\theta$. □

2.3. Inherited properties

The quotient inherits the dynamical properties of the doubling map:

Property on S^1	**Inherited property on** $[0, 1]$
Lebesgue measure $\frac{d\theta}{2\pi}$	Arcsine measure $\mu_4 = \frac{dx}{\pi\sqrt{x(1-x)}}$
Topological entropy $\log 2$	Topological entropy $\log 2$
Full shift on $\{0, 1\}^{\mathbb{N}}$	Full shift on $\{0, 1\}^{\mathbb{N}}$
2-to-1 covering	2-to-1 on $(0, 1)$, with fold at $x = \frac{1}{2}$

Table 1: Dynamical properties inherited by the quotient.

The invariant measure μ_4 is the pushforward of normalized Lebesgue measure under $x = \sin^2(\theta/2)$. By the change-of-variables formula, $d\theta = \frac{dx}{\sqrt{x(1-x)}}$, and normalization over $[0, 1]$ introduces the factor $1/\pi$.

Remark. The semiconjugacy $h(\theta) = \sin^2(\theta/2)$ satisfies $h \circ D = f_4 \circ h$, and h is exactly 2-to-1 except at the fixed points $\theta = 0$ and $\theta = \pi$ of β. The map f_4 is the unique quadratic polynomial semiconjugate to the doubling map via a quotient by an involution commuting with

D. The coefficient $r = 4$ is forced: it equals $\left(\max_{[0,1]} x(1-x)\right)^{-1} = (1/4)^{-1} = 4$, ensuring that $f_4([0,1]) = [0,1]$.

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3. The Klein Four-Group

3.1. Symmetries of the doubling map

The doubling map $D(z) = z^2$ on $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ admits a group of four symmetries:

$$V_4 = \{e, \alpha, \beta, \alpha\beta\},$$

where

- $\alpha : z \mapsto -z$ (the **deck transformation**: $D(\alpha(z)) = D(z)$),
- $\beta : z \mapsto z^{-1}$ (the **commuting involution**: $D(\beta(z)) = \beta(D(z))$; agrees with complex conjugation on S^1),
- $\alpha\beta : z \mapsto -z^{-1}$ (the composition).

These satisfy $\alpha^2 = \beta^2 = (\alpha\beta)^2 = e$ and $\alpha\beta = \beta\alpha$, so $V_4 \cong \mathbb{Z}/2 \times \mathbb{Z}/2$. On the circle S^1, the actions in angular coordinates are:

Element	Action on θ	Geometric meaning
e	θ	Identity
α	$\theta + \pi$	Antipodal map
β	$-\theta$	Reflection
$\alpha\beta$	$\pi - \theta$	Complementary reflection

Table 2: Actions of V_4 in angular coordinates on S^1.

Remark. The symmetry group is exactly V_4 and no larger. The deck group of D has order $\deg(D) = 2$, contributing one $\mathbb{Z}/2$ factor. The unique non-trivial automorphism of S^1 commuting with D and compatible with the real structure is β, contributing the other. The product $2 \times 2 = 4 = |V_4|$ is thus forced by the degree.

3.2. Subgroups and quotients

The Klein four-group has exactly three subgroups of order 2. Two of them yield clean quotient dynamical systems:

Quotient by $\langle\beta\rangle$ ($z \sim z^{-1}$, i.e., $\theta \sim -\theta$). The quotient space is $[0,1]$ and the descended map is the **logistic map** $f_4(x) = 4x(1-x)$, as shown in Section 2.

Quotient by $\langle\alpha\rangle$ ($z \sim -z$, i.e., $\theta \sim \theta + \pi$). The quotient space is again a circle S^1, parameterized by $w = z^2$, and the descended map is $w \mapsto w^2$ — the **doubling map** again (or equivalently, the **power map** on the quotient circle).

Both descended systems have topological entropy $\log 2$, full shift symbolic dynamics, and equivalent invariant measures.

> **Remark.** The third involution $\alpha\beta : \theta \mapsto \pi - \theta$ does *not* commute with D, nor is it a deck transformation; one checks that $D\circ(\alpha\beta) = \beta\circ D \neq (\alpha\beta)\circ D$. Consequently, D does not descend to a well-defined map on $S^1/\langle\alpha\beta\rangle$. However, the identity $D\circ(\alpha\beta) = \beta\circ D$ shows that D intertwines the $\alpha\beta$- and β-quotients. On the β-quotient $[0, 1]$ (parameterized by $x = \sin^2(\theta/2)$), the homeomorphism $\varphi(x) = (2/\pi)\arcsin\left(\sqrt{x}\right)$ of $[0, 1]$ conjugates f_4 to the tent map $T(x) = 1 - |2x - 1|$, i.e., $\varphi\circ f_4 = T\circ\varphi$. This conjugacy straightens the arcsine measure to Lebesgue measure.

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4. The Harmonic Range Characterization

4.1. The four structural points

For $r \in (2, 4]$, the quadratic map $f_r(z) = rz(1 - z)$, viewed as a rational map of degree 2 on $\mathbb{CP}^1$, has four dynamically distinguished points:

Point	Value	Role
c	$1/2$	$f_r'(c) = 0$ (critical point)
v	$r/4$	$f_r(c)$ (critical value)
p	$1 - 1/r$	Non-trivial fixed point (multiplier $f_r'(p) = 2 - r$)
∞	∞	Superattracting fixed point ($f_r(\infty) = \infty$, local degree 2)

Table 3: The four dynamically distinguished points of f_r.

These four points are distinct for $r \in (2, 4]$ (at $r = 2$, we have $c = p = 1/2$). Their configuration on the projective line $\mathbb{CP}^1$ is governed by the cross-ratio.

4.2. The cross-ratio

Using the standard convention for cross-ratio with one point at infinity:

$$(c, v; p, \infty) = \frac{c - p}{v - p}.$$

Substituting the values:

$$
\begin{aligned}
\lambda(r) &= \frac{1/2 - (1 - 1/r)}{r/4 - (1 - 1/r)} \\
&= \frac{(2-r)/(2r)}{(r-2)^2/(4r)} \\
&= \frac{4(2-r)}{2(r-2)^2} \\
&= \frac{-2}{r-2}.
\end{aligned}
$$

Thus:

$$\boxed{\lambda(r) = (c, v; p, \infty) = \tfrac{-2}{r-2}.}$$

This is a Möbius function of r, taking the values:

r	$\lambda(r)$	**Configuration**
2	$\pm\infty$	Degenerate ($c = p$)
3	-2	Generic
4	-1	**Harmonic**

Table 4: Cross-ratio values at distinguished parameters.

4.3. The characterization

Proposition (*4.3*). *Among the logistic maps f_r with $r \in (2, 4]$, the four structural points $(c, v; p, \infty)$ form a harmonic range — that is, their cross-ratio equals -1 — if and only if $r = 4$.*

Proof. Setting $\lambda(r) = -1$:

$$\frac{-2}{r-2} = -1 \Longrightarrow r - 2 = 2 \Longrightarrow r = 4.$$

■

Corollary. *At $r = 4$, the non-trivial fixed point is the affine midpoint of the critical point and the critical value:*

$$p = \frac{c+v}{2} = \frac{1/2 + 1}{2} = \frac{3}{4}.$$

This is equivalent to the harmonic range condition when one of the four points is ∞.

Proof. When $D = \infty$, the cross-ratio $(A, B; C, \infty) = \frac{A-C}{B-C}$ equals -1 if and only if $C = \frac{A+B}{2}$. ■

4.4. The other fixed point

The logistic map f_r has *two* finite fixed points: the non-trivial fixed point $p = 1 - 1/r$ used above, and the trivial fixed point $q = 0$. One may ask whether the harmonic geometry depends on this choice.

Computing the cross-ratio with q in place of p:

$$(c, v; q, \infty) = \frac{c - q}{v - q} = \frac{1/2 - 0}{r/4 - 0} = \frac{2}{r}.$$

At $r = 4$, this evaluates to $1/2$. The anharmonic orbit of the harmonic cross-ratio -1 is

$$\{-1, 1/2, 2\},$$

obtained by applying the six Möbius transformations $\lambda \mapsto \{\lambda, 1/\lambda, 1 - \lambda, 1/(1 - \lambda), \lambda/(\lambda - 1), (\lambda - 1)/\lambda\}$ to $\lambda = -1$ (see, e.g., [4]). Since $1/2$ belongs to this orbit, the four points $(c, v; q, \infty)$ also form a harmonic range at $r = 4$.

This is not coincidental. The cross-ratio $(c, v; q, \infty) = 1/2$ lies in the same anharmonic orbit as $(c, v; p, \infty) = -1$: the substitution of q for p corresponds to a transposition in S_4 that changes the cross-ratio from -1 to $1/2$ via the Möbius transformation $\lambda \mapsto \lambda/(\lambda - 1)$. Indeed, $(-1)/((-1) - 1) = (-1)/(-2) = 1/2$.

More formally: when the fourth point is fixed at $D = \infty$, the remaining three points may be permuted freely; this residual symmetry is the group S_3 of order 6, which acts on the cross-ratio by exactly the six Möbius transformations listed above. The anharmonic orbit $\{-1, 1/2, 2\}$ is precisely the S_3-orbit of any harmonic cross-ratio value. Substituting p for q permutes the arguments $(c, v; \cdot, \infty)$, thereby moving within this orbit.

The harmonic geometry of f_4 is therefore *robust*: it holds regardless of which fixed point — p or q — is chosen as the third element of the cross-ratio tuple.

4.5. The harmonic involution

The classical characterization of the harmonic range involves involutions. Given four points on $\mathbb{CP}^1$, there are two distinct types of associated Möbius involutions:

1. **Pair-swapping involutions.** For any four distinct points $(A, B; C, D)$, there exists a Möbius involution simultaneously swapping $A \leftrightarrow B$ and $C \leftrightarrow D$. This exists for *all* four-point configurations, regardless of the cross-ratio.
2. **Fixed-point involutions.** The Möbius involution with *fixed points* C and D maps A to B if and only if $(A, B; C, D) = -1$.

It is the second type that characterizes the harmonic range. At $r = 4$, the unique Möbius involution with fixed points $p = 3/4$ and ∞ is the affine reflection about p:

$$S(z) = 3/2 - z,$$

and indeed $S(c) = S(1/2) = 1 = v$ and $S(v) = S(1) = 1/2 = c$. The existence of this involution — fixing the pair $\{p, \infty\}$ while exchanging the pair $\{c, v\}$ — is equivalent to the harmonic condition $\lambda = -1$ and therefore to $r = 4$.

For $r \neq 4$, the affine reflection about $p = 1 - 1/r$ is $S_r(z) = 2(1 - 1/r) - z$, and one computes $S_r(c) = S_r(1/2) = 3/2 - 2/r$. This equals $v = r/4$ only when $3/2 - 2/r = r/4$, which gives $r^2 - 6r + 8 = (r-2)(r-4) = 0$. Since $r > 2$, we obtain $r = 4$.

There also exists, for any $r \in (2, 4]$, a pair-swapping involution T_r that simultaneously exchanges $\{c, v\}$ and $\{p, \infty\}$. At $r = 4$, this takes the explicit form:

$$T(z) = \frac{6z - 5}{8z - 6},$$

which satisfies $T^2 = \mathrm{id}$, $T(1/2) = 1$, $T(1) = 1/2$, $T(3/4) = \infty$, $T(\infty) = 3/4$.

Verification. We need T to be a Möbius transformation of order 2 with $T(1/2) = 1$, $T(1) = 1/2$, $T(3/4) = \infty$, and $T(\infty) = 3/4$.

From $T(\infty) = 3/4$, write $T(z) = \frac{az+b}{cz+d}$; then $T(\infty) = a/c = 3/4$, so set $a = 3$, $c = 4$: $T(z) = \frac{3z+b}{4z+d}$.

From $T(3/4) = \infty$: $4(3/4) + d = 0$, giving $d = -3$. So $T(z) = \frac{3z+b}{4z-3}$.

From $T(1/2) = 1$: $\frac{3/2+b}{2-3} = 1$, giving $3/2 + b = -1$, so $b = -5/2$. Hence $T(z) = \frac{3z-5/2}{4z-3} = \frac{6z-5}{8z-6}$.

Check: $T(1) = \frac{6-5}{8-6} = \frac{1}{2}$. ✓

Check:

$$\begin{aligned} T^2(z) = T\left(\frac{6z-5}{8z-6}\right) &= \frac{6 \cdot \frac{6z-5}{8z-6} - 5}{8 \cdot \frac{6z-5}{8z-6} - 6} \\ &= \frac{36z - 30 - 5(8z-6)}{48z - 40 - 6(8z-6)} \\ &= \frac{36z - 30 - 40z + 30}{48z - 40 - 48z + 36} \\ &= \frac{-4z}{-4} = z. \quad \checkmark \end{aligned}$$

The distinction between these two types of involution is essential: the pair-swapping involution T exists for all r and carries no special information, while the fixed-point involution S exchanging c and v exists only at $r = 4$ and encodes the harmonic condition.

4.6. Connection to V_4

The link between the harmonic range and the Klein four-group is classical (see, e.g., [4]). Given four ordered points (z_1, z_2, z_3, z_4) on $\mathbb{CP}^1$, the symmetric group S_4 acts by permutation, and the cross-ratio transforms under a homomorphism $S_4 \to S_3$ to the group of anharmonic ratios $\{\lambda, 1/\lambda, 1-\lambda, 1/(1-\lambda), \lambda/(\lambda-1), (\lambda-1)/\lambda\}$. The kernel of this homomorphism is the Klein four-group

$$V_4 = \{e, (12)(34), (13)(24), (14)(23)\} \triangleleft S_4,$$

consisting of the double transpositions. These are precisely the permutations that preserve the cross-ratio for *all* values of λ.

At $\lambda = -1$, the anharmonic orbit $\{-1, 2, 1/2\}$ has only three elements instead of the generic six, meaning the stabilizer of the harmonic cross-ratio in S_4 is a group of order $24/3 = 8$. This is the dihedral group D_4 (the symmetry group of a square), which contains V_4 as a normal subgroup of index

2. The extra symmetry at $\lambda = -1$ corresponds to the transpositions that

exchange the two pairs within the harmonic configuration — precisely the enhanced symmetry that distinguishes the harmonic range from a generic four-point configuration.

Thus the Klein four-group appears in two roles simultaneously: as the universal stabilizer of the cross-ratio (for any λ), and as a subgroup of the *enhanced* stabilizer (of order 8) that is specific to the harmonic value $\lambda = -1$. Both roles are present at $r = 4$.

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5. The Belyi Property and the Moduli Space $\mathscr{M}_{0,4}$

5.1. Belyi maps and the logistic family

Definition. A rational map $f : \mathbb{CP}^1 \to \mathbb{CP}^1$ is a **Belyi map** if its critical values lie in $\{0, 1, \infty\}$.

Theorem (*5.1*). *Among the logistic maps $f_r(z) = rz(1-z)$ with $r \in \mathbb{C}$, the following are equivalent:*

1. *f_r is a Belyi map.*
2. *f_r is postcritically finite with postcritical set $\mathrm{PC}(f_r) \subseteq \{0, 1, \infty\}$.*
3. *$r \in \{0, 4\}$.*

The case $r = 0$ is trivial (constant map). The case $r = 4$ is the unique non-trivial Belyi map in the family.

Proof. By Riemann–Hurwitz, f_r has exactly two critical points on $\mathbb{CP}^1$: the finite critical point $c = 1/2$ (with critical value $v = r/4$) and the point ∞ (with $f_r(\infty) = \infty$, local degree 2). Since $\infty \in \{0, 1, \infty\}$, the Belyi condition reduces to $v = r/4 \in \{0, 1, \infty\}$:

- $r/4 = 0 \Rightarrow r = 0$.
- $r/4 = 1 \Rightarrow r = 4$.
- $r/4 = \infty$: impossible for finite r.

For (2) $\Rightarrow$ (1): the postcritical set $\mathrm{PC}(f)$ is defined as the closure of the union of forward orbits of all critical values, so in particular $\mathrm{PC}(f) \supseteq \{\text{critical values}\}$. If $\mathrm{PC}(f) \subseteq \{0, 1, \infty\}$, then all critical values lie in $\{0, 1, \infty\}$, which is the Belyi condition.

At $r = 4$, postcritical finiteness follows from the forward orbit of the finite critical value: $1 \xmapsto{f_4} 0 \xmapsto{f_4} 0$, so $\mathrm{PC}(f_4) = \{0, 1\} \cup \{\infty\} = \{0, 1, \infty\}$. ∎

5.2. Arithmetic significance

The Belyi property explains *why* the harmonic configuration is arithmetically special. By Belyi's theorem [5], a smooth projective curve over $\mathbb{C}$ is defined over $\overline{\mathbb{Q}}$ if and only if it admits a Belyi map. The map $f_4 : \mathbb{CP}^1 \to \mathbb{CP}^1$ is itself a Belyi map defined over $\mathbb{Q}$, and its *dessin d'enfants* — the preimage $f_4^{-1}([0, 1])$ — is the interval $[0, 1]$ with vertices at 0, $1/2$, and 1: a tree with two edges, the simplest non-trivial dessin of degree 2.

The connection to the harmonic range is direct: the four structural points $(c, v; p, \infty) = (1/2, r/4; 1 - 1/r, \infty)$ form a harmonic range when $v = r/4 = 1$, which is precisely the Belyi condition $v \in \{0, 1, \infty\}$ (taking the non-trivial case $v = 1$). The harmonic range condition and the Belyi condition are not merely equivalent — they are the *same* constraint, expressed in projective-geometric and arithmetic language respectively. For further background on arithmetic dynamics, see [6].

5.3. Postcritical finiteness and Thurston rigidity

By Thurston's rigidity theorem [7], postcritically finite rational maps are determined up to Möbius conjugacy by their combinatorial data (the *Hubbard tree* or, equivalently, the homotopy class of the branched covering relative to the postcritical set). The map f_4 is the unique quadratic polynomial (up to affine conjugacy) whose critical point has orbit length 3 and lands on a fixed point.

The harmonic condition is intrinsic to the conjugacy class, not the particular affine model. The logistic map $f_4(x) = 4x(1 - x)$ is affinely conjugate to $g(z) = z^2 - 2$ via $h(x) = 2 - 4x$, so that $h \circ f_4 = g \circ h$. Since cross-ratios are Möbius-invariant, the structural points of g — critical point 0, critical value -2, non-trivial fixed point -1, and ∞ — form the same harmonic configuration:

$$(0, -2; -1, \infty) = \frac{0 - (-1)}{-2 - (-1)} = \frac{1}{-1} = -1.$$

Using the trivial fixed point 2, one obtains $(0, -2; 2, \infty) = 1/2$, which lies in the anharmonic orbit $\{-1, 1/2, 2\}$, consistent with Section 4.4.

> **Remark.** The map $g(z) = z^2 - 2$ is the degree-2 Chebyshev polynomial T_2 and represents the core PCF map of the real quadratic family at the tip of the Mandelbrot set. The harmonic condition thus characterizes this distinguished dynamical system in a coordinate-independent manner.

5.4. The moduli space $\mathscr{M}_{0,4}$

The moduli space of four distinct ordered points on $\mathbb{CP}^1$, up to Möbius equivalence, is

$$\mathscr{M}_{0,4} \cong \mathbb{CP}^1 \setminus \{0, 1, \infty\},$$

parameterized by the cross-ratio λ. Its Deligne–Mumford compactification is $\overline{\mathscr{M}}_{0,4} \cong \mathbb{CP}^1$, with boundary divisor $\partial\overline{\mathscr{M}}_{0,4} = \{0, 1, \infty\}$ — the locus where two of the four points collide.

The postcritical set of f_4 satisfies

$$\mathrm{PC}(f_4) = \{0, 1, \infty\} = \partial\overline{\mathscr{M}}_{0,4}.$$

This identity is a consequence of the Belyi property: a Belyi map is precisely a map whose ramification is supported on $\partial\overline{\mathscr{M}}_{0,4}$. The maximally entropic degree-2 polynomial has its postcritical set equal to the boundary of the moduli space parameterizing four-point configurations on $\mathbb{CP}^1$; this identity is equivalent to (and clarified by) the Belyi property, and the Belyi condition is equivalent to the harmonic range condition on the four structural points.

We note that the number 4 — the number of marked points — is the minimum needed for a non-trivial cross-ratio (three points on $\mathbb{CP}^1$ can always be sent to $\{0, 1, \infty\}$ by a Möbius transformation), so the logistic map operates at the threshold where projective-geometric invariants first become available.

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6. Entropy and Surjectivity

6.1. Surjectivity and the full shift

The connection between $r = 4$ and maximal entropy $\log 2$ is mediated by surjectivity. For $r < 4$, the critical value $v = r/4 < 1$, so $f_r([0, 1]) = [0, r/4] \subsetneq [0, 1]$. Points in $(r/4, 1]$ have no preimage, certain symbolic sequences are forbidden, and the shift space is a proper subshift with entropy strictly less than $\log 2$.

At $r = 4$, the critical value $v = 1$, so $f_4([0, 1]) = [0, 1]$. Every point has a preimage, every binary itinerary is realized, and the symbolic dynamics is the full shift on $\{0, 1\}^{\mathbb{N}}$ with entropy $\log 2$.

For $r > 4$, the critical value $v > 1$ and some orbits escape to $+\infty$. The non-escaping set is a Cantor set of Lebesgue measure zero on which f_r is again conjugate to the full shift, but the invariant set no longer fills the interval.

Thus $r = 4$ is the unique value where the entropy is maximal *and* the invariant set has full Lebesgue measure.

6.2. The variational principle

The variational principle for topological entropy [8] states

$$h_{\mathrm{top}}(f) = \sup_{\mu \in \mathscr{M}(f)} h_\mu(f),$$

where $\mathscr{M}(f)$ is the space of f-invariant Borel probability measures and h_μ denotes the measure-theoretic entropy. For the logistic family:

- Topological entropy $h_{\text{top}}(f_r)$ increases monotonically from 0 to $\log 2$ as r increases from 1 to 4 (see [2], [9]). In particular, $h_{\text{top}}(f_r) < \log 2$ strictly for $r < 4$.
- For each r, there exists a unique measure of maximal entropy μ_r achieving the supremum; this is the measure maximizing entropy at the (strictly lower) level $h_{\text{top}}(f_r)$.
- For $r < 4$, the unique measure of maximal entropy μ_r is singular with respect to Lebesgue measure.
- At $r = 4$, the unique measure of maximal entropy $\mu_4 = \frac{dx}{\pi\sqrt{x(1-x)}}$ is absolutely continuous with respect to Lebesgue measure.

The transition from singular to absolutely continuous at $r = 4$ represents a qualitative change in the space of invariant measures, occurring precisely at the harmonic range configuration. This transition concerns the *measure of maximal entropy* for each parameter r; the nature of this particular measure changes qualitatively as r increases through 4. (Jakobson's theorem [10] guarantees absolutely continuous invariant measures for a positive-measure set of parameters $r < 4$, but those measures are *not* the measures of maximal entropy for their respective parameters.)

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7. Synthesis and Concluding Remarks

7.1. Summary of equivalences

The following conditions on the parameter $r \in (2, 4]$ are equivalent:

1. $f_r : [0, 1] \to [0, 1]$ is surjective.
2. The topological entropy $h_{\text{top}}(f_r) = \log 2$.
3. The measure of maximal entropy is absolutely continuous with respect to Lebesgue measure on $[0, 1]$.
4. The critical orbit is finite: $f_r(v) = 0$, i.e., the critical value maps to the fixed point 0 in one step.
5. The cross-ratio $(c, v; p, \infty) = -1$ (harmonic range).
6. f_r is a Belyi map (critical values in $\{0, 1, \infty\}$).
7. The Möbius involution fixing p and ∞ exchanges c and v.

Each condition is equivalent to $r = 4$. The equivalence of (1)–(4) is well established in the dynamical systems literature (see [2], [3], [11]). The equivalence of (5) with the others is the content of Proposition 4.3. The equivalence of (6) is established in Theorem 5.1. The equivalence of (7) with (5) is the classical characterization of the harmonic range via fixed-point involutions (Section 4.5).

These seven conditions belong to four distinct mathematical domains — dynamics (1–3), combinatorics of orbits (4), projective geometry (5, 7), and arithmetic geometry (6) — yet they select

the same unique parameter. This confluence reflects the rigidity of postcritically finite maps at the boundary of the Mandelbrot set.

7.2. The role of V_4

The Klein four-group $V_4 \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ appears throughout this story in three interrelated roles:

1. **Dynamical symmetry.** V_4 is the symmetry group of the doubling map $D(z) = z^2$ on S^1, generated by the deck transformation $\alpha : z \mapsto -z$ and the commuting involution $\beta : z \mapsto \overline{z}$.
2. **Cross-ratio stabilizer.** V_4 is the kernel of the homomorphism $S_4 \to S_3$ given by the action of permutations on the cross-ratio. It is the universal group of permutations that preserve the cross-ratio, and it sits inside the order-8 stabilizer of the harmonic cross-ratio $\lambda = -1$.
3. **Harmonic symmetry.** At $r = 4$, the three non-trivial involutions of $\mathbb{CP}^1$ that permute the four structural points by double transpositions — the pair-swapping involution T of Section 4.5, together with the involutions swapping $\{c, p\}$ with $\{v, \infty\}$ and $\{c, \infty\}$ with $\{v, p\}$ — realize the abstract V_4 acting on the harmonic configuration. This V_4 action on the set of four points is distinct from the dynamical V_4 of role (1), but both originate from the degree-2 structure of the underlying map.

These three incarnations of V_4 are unified by the quotient construction: the logistic map at $r = 4$ *is* the doubling map modulo one factor of V_4, and the harmonic range condition *is* the projective-geometric signature of this quotient achieving surjectivity.

7.3. Structural origins

Every appearance of the number 4 in this context traces to the degree of the doubling map:

- $r = 4 = \left(\max_{[0,1]} x(1-x)\right)^{-1} = (1/4)^{-1}$, forced by surjectivity.
- $|V_4| = 4 = |\operatorname{Aut}_D(S^1)| = (\deg D) \times |\{\text{commuting involutions}\}| = 2 \times 2$.
- A degree-2 polynomial on $\mathbb{CP}^1$ has exactly one finite critical point (Riemann–Hurwitz), one critical value, and two finite fixed points. Together with the point ∞ (which is simultaneously a fixed point and a critical point), these yield 4 dynamically distinguished points — precisely the minimum needed for a non-trivial cross-ratio.
- The harmonic range is the unique non-degenerate four-point configuration with enhanced D_4 symmetry of order $2|V_4| = 8$.

The number $4 = 2^2$ thus arises as the square of the degree, or equivalently as the order of the product of two independent $\mathbb{Z}/2$ symmetries inherent in any degree-2 map with a real structure.

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